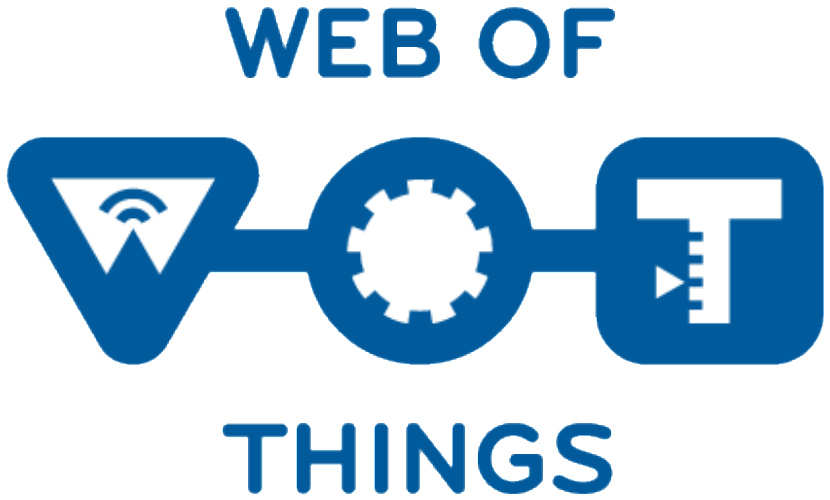


Media Streaming Use Cases in WoT

W3C TPAC2026 breakout – Web of Things Live Demonstration

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Overview

- WoT Thing Description (TD) can describes interactions as "properties", "actions", and "events"
- These assume a request-response pattern, or a connection per request with continuous responses
- Media streaming needs different patterns:
  - Data exchanged over a separate connection from the request
  - Multiple tracks (audio, video, data) in one connection
  - Sending and receiving data over the same connection
  - Control of an established connection (stop, resume)
- Plugfest case study: MediaMTX[1], software that relays various media streams

```
{
    "@context": [
        "https://www.w3.org/2022/wot/td/v1.1",
        { "saref": "https://w3id.org/saref#" }
    ],
    "id": "urn:uuid:300f4c4b-ca6b-484a-88cf-fd5224a9a61d",
    "title": "MyLampThing",
    "@type": "saref:LightSwitch",
    "securityDefinitions": {
        "basic_sc": {"scheme": "basic", "in": "header"}
    },
    "security": "basic_sc",
    "properties": {
        "status": {
            "@type": "saref:OnOffState",
            "type": "string",
            "forms": [{
                "href": "https://mylamp.example.com/status"
            }]
        }
    },
    "actions": {
        "toggle": {
            "@type": "saref:ToggleCommand",
            "forms": [{
                "href": "https://mylamp.example.com/toggle"
            }]
        }
    },
    "events": {
        "overheating": {
            "data": {"type": "string"},
            "forms": [{
                "href": "https://mylamp.example.com/oh"
            }]
        }
    }
}
```

Property

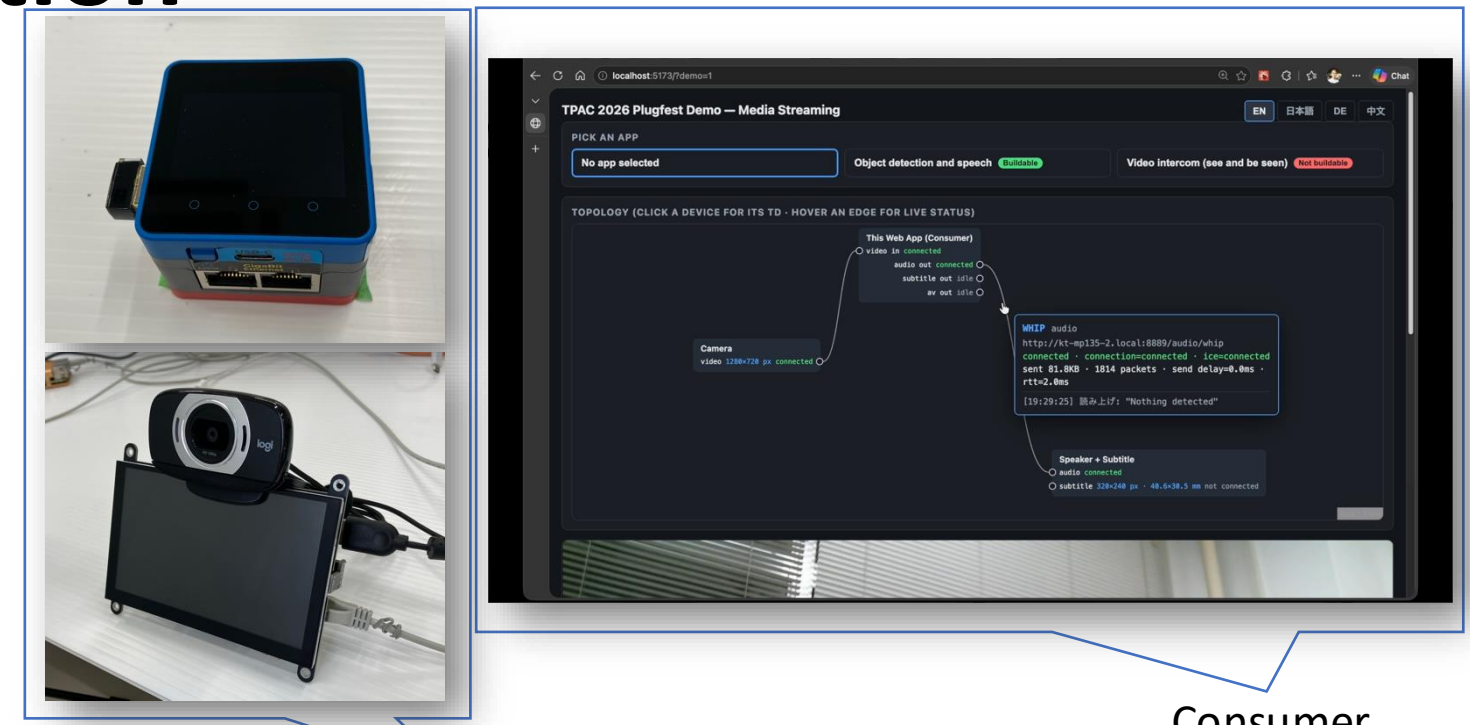
Action

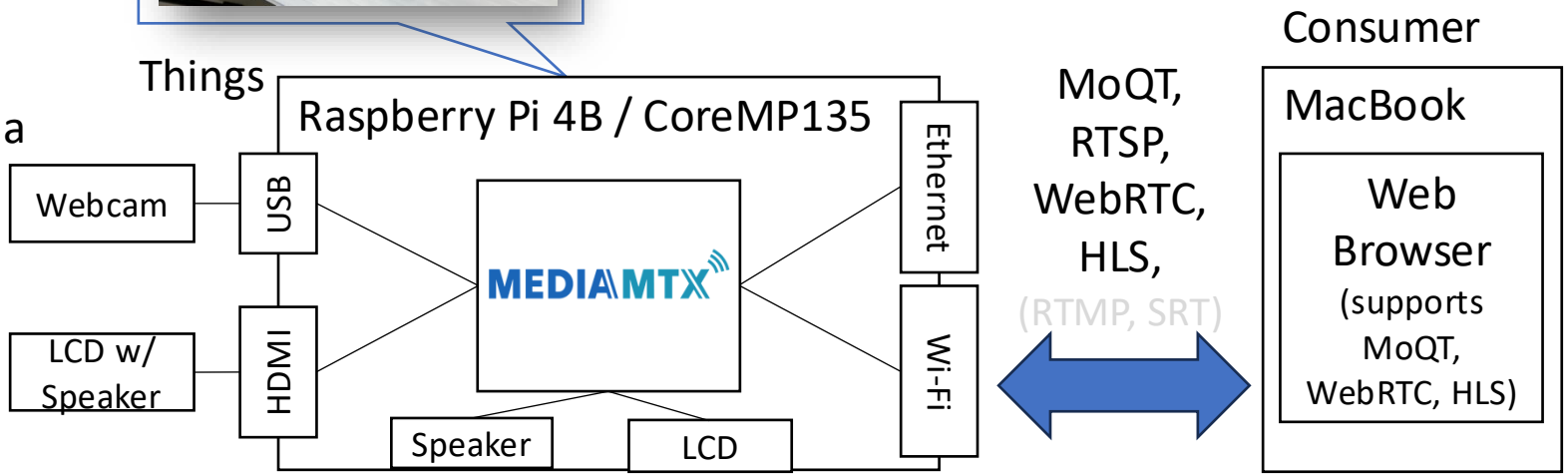
Event

[1] MediaMTX: ready-to-use, open source, live media router, https://mediamtx.org/

Plugfest Configuration

- Things: send and receive video and audio using MediaMTX
  - e.g., send camera video, show received video on a display
  - Supported protocols: MoQT, RTSP, RTMP, WebRTC, SRT, HLS
- Consumer: browser application
  - Collects TDs of nearby Things and combines them into apps, such as object detection with audio/video output, or video calling
  - Finds endpoints usable in a browser (MoQT, WebRTC, HLS) and recommends a configuration that meets the app's latency requirements
- Question: what information should a TD contain to enable such a Consumer?





Demo



Lessons Learned

What we found:

- A single-track, unidirectional media stream can be described as a property
- A URI alone is not enough to choose the right form (i.e. endpoint, protocol)
  - WHIP, WHEP, and HLS forms all use http(s) URIs
  - Distinguish them by contentType or subprotocol, or by additional vocabulary
- Encoding, resolution, fps, and latency also help select an endpoint
  - External vocabularies could be reused: Media Capabilities, WebCodecs Codec Registry
- How to handle a connection with multiple tracks is still open
  - Option 1: One property per track, plus a manageable affordance for creating and deleting the connection, linked in the TD
  - Option 2: One property with audio and video tracks, with connection create/delete described in its forms
  - Option 3: Other ?

Open issues:

- Bidirectional streaming
- Data tracks
- Stream control